

District 4 Model Comparison - Geospatial Data: 2023 - 2025 Sales

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Model I

Table 1: Model Metrics: LN-SPPSF Model Comparison

Metric Source	n	df_resid	df_model	r	R^2	Adj. R^2	SEE	f_stat	AIC	BIC
Control	1298	1286	11	0.3145	0.0989	0.0912	0.4876	12.8330	1833	1900
+ Geospatial	2263	2244	18	0.3522	0.1240	0.1170	0.5179	17.6495	3464	3579
% Difference	74.35%	74.49%	63.64%	11.97%	25.38%	28.27%	6.20%	37.53%	88.98%	88.37%

Table 2: Ratio Metrics: LN-SPPSF Model Comparison

Metric Source	median_ratio	mean_ratio	wmean_ratio	PRD	COD	PRB	Within 10%	Within 20%	Within 50%
Control	0.9809	1.1261	0.8880	1.2681	43.6134	-1.0215	15.95%	30.28%	72.19%
+ Geospatial	0.9887	1.1436	0.8787	1.3014	46.5667	-0.9849	14.32%	27.88%	68.14%
% Difference	0.79%	1.55%	-1.05%	2.62%	6.77%	3.58%	-10.22%	-7.91%	-5.61%

Model II

Table 3: Model Metrics: LN-PRICE (No ECF) Model Comparison

Metric Source	n	df_resid	df_model	r	R^2	Adj. R^2	SEE	f_stat	AIC	BIC
Control	1708	1688	19	0.5202	0.2706	0.2624	0.5020	32.9646	2514	2628
+ Geospatial	2263	2241	21	0.5161	0.2663	0.2594	0.5147	38.7345	3439	3571
% Difference	32.49%	32.76%	10.53%	-0.80%	-1.60%	-1.14%	2.53%	17.50%	36.79%	35.88%

Table 4: Ratio Metrics: LN-PRICE (No ECF) Model Comparison

Metric Source	median_ratio	mean_ratio	wmean_ratio	PRD	COD	PRB	Within 10%	Within 20%	Within 50%
Control	0.9820	1.1333	0.8853	1.2802	45.1230	-0.5993	15.46%	28.98%	69.85%
+ Geospatial	1.0035	1.1406	0.8802	1.2958	45.3313	-0.6031	14.63%	28.50%	68.40%
% Difference	2.19%	0.64%	-0.57%	1.22%	0.46%	-0.63%	-5.37%	-1.65%	-2.07%

Model III

Table 5: Model Metrics: LN-PRICE Model Comparison

Metric Source	n	df_resid	df_model	r	R^2	Adj. R^2	SEE	f_stat	AIC	BIC
Control	1708	1676	31	0.6399	0.4094	0.3985	0.4533	37.4794	2178	2357
+ Geospatial	2263	2233	29	0.6304	0.3974	0.3896	0.4672	50.7869	3010	3187
% Difference	32.49%	33.23%	-6.45%	-1.47%	-2.93%	-2.23%	3.08%	35.51%	38.20%	35.21%

Table 6: Ratio Metrics: LN-PRICE Model Comparison

Metric Source	median_ratio	mean_ratio	wmean_ratio	PRD	COD	PRB	Within 10%	Within 20%	Within 50%
Control	0.9842	1.1048	0.9079	1.2168	39.8082	-0.4030	16.16%	31.03%	73.54%
+ Geospatial	0.9918	1.1122	0.9027	1.2321	41.1634	-0.4199	14.71%	28.77%	72.47%
% Difference	0.78%	0.67%	-0.58%	1.26%	3.40%	-4.18%	-8.94%	-7.29%	-1.45%

Model IV

Table 7: Model Metrics: LN-PRICE (Inliers) Model Comparison

Metric Source	n	df_resid	df_model	r	R^2	Adj. R^2	SEE	f_stat	AIC	BIC
Control	1577	1547	29	0.6938	0.4814	0.4717	0.3945	49.5152	1573	1739
+ Geospatial	2128	2098	29	0.6751	0.4557	0.4482	0.4141	60.5730	2318	2494
% Difference	34.94%	35.62%	0.00%	-2.70%	-5.33%	-4.98%	4.98%	22.33%	47.36%	43.42%

Table 8: Ratio Metrics: LN-PRICE (Inliers) Model Comparison

Metric Source	median_ratio	mean_ratio	wmean_ratio	PRD	COD	PRB	Within 10%	Within 20%	Within 50%
Control	0.9824	1.0787	0.9300	1.1600	34.6513	-0.3122	18.52%	34.81%	78.50%
+ Geospatial	0.9889	1.0876	0.9219	1.1797	36.7374	-0.3360	16.31%	31.53%	76.64%
% Difference	0.66%	0.82%	-0.87%	1.70%	6.02%	-7.63%	-11.93%	-9.42%	-2.37%

Model V

Table 9: Model Metrics: LN-TASP Model Comparison

Metric Source	n	df_resid	df_model	r	R^2	Adj. R^2	SEE	f_stat	AIC	BIC
Control	1577	1548	28	0.6936	0.4810	0.4717	0.3944	51.2463	1571	1732
+ Geospatial	2128	2099	28	0.6727	0.4526	0.4453	0.4141	61.9778	2317	2486
% Difference	34.94%	35.59%	0.00%	-3.00%	-5.92%	-5.59%	4.99%	20.94%	47.49%	43.53%

Table 10: Ratio Metrics: LN-TASP Model Comparison

Metric Source	median_ratio	mean_ratio	wmean_ratio	PRD	COD	PRB	Within 10%	Within 20%	Within 50%
Control	0.9824	1.0788	0.9302	1.1597	34.6511	-0.3125	18.45%	35.07%	78.38%
+ Geospatial	0.9883	1.0876	0.9220	1.1796	36.7698	-0.3404	16.40%	31.77%	76.60%
% Difference	0.60%	0.82%	-0.88%	1.71%	6.11%	-8.96%	-11.12%	-9.41%	-2.27%

Model VI

Table 11: Model Metrics: LN-TASP Model II Comparison

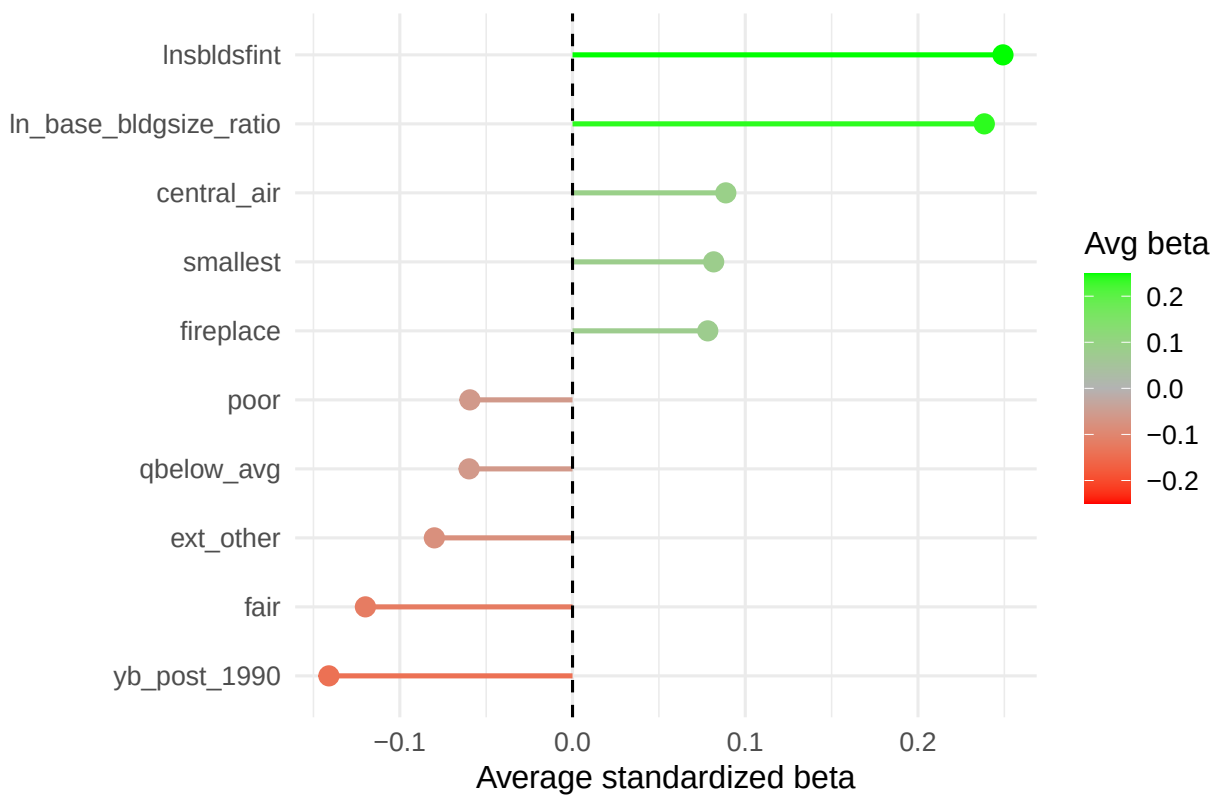
Metric Source	n	df_resid	df_model	r	R^2	Adj. R^2	SEE	f_stat	AIC	BIC
Control	1577	1549	27	0.6937	0.4812	0.4721	0.3942	53.2048	1568	1724
+ Geospatial	2128	2095	32	0.6753	0.4560	0.4477	0.4132	54.8703	2311	2504
% Difference	34.94%	35.25%	18.52%	-2.65%	-5.24%	-5.18%	4.81%	3.13%	47.39%	45.24%

Table 12: Ratio Metrics: LN-TASP Model II Comparison

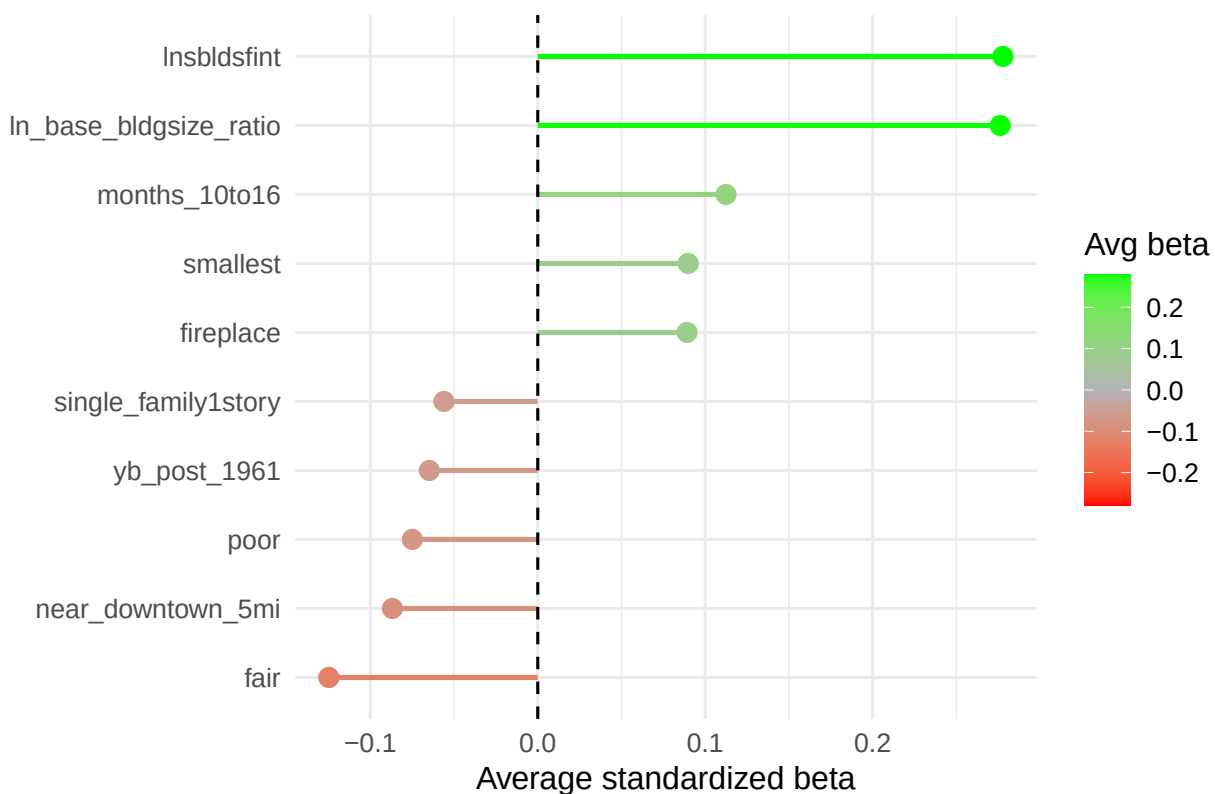
Metric Source	median_ratio	mean_ratio	wmean_ratio	PRD	COD	PRB	Within 10%	Within 20%	Within 50%
Control	0.9879	1.0787	0.9300	1.1599	34.3950	-0.3113	18.83%	35.26%	78.69%
+ Geospatial	0.9946	1.0869	0.9224	1.1783	36.4125	-0.3345	16.49%	31.81%	76.74%
% Difference	0.68%	0.76%	-0.81%	1.58%	5.87%	-7.45%	-12.42%	-9.77%	-2.48%

Model Comparison

District 4 | Avg Std Beta across Initial Models



District 4 | Avg Std Beta across Geospatial Models



Prediction Testing

Table 13: Deviation Summary: Initial LN-SPPSF Model

Metric	Value
Average sale price deviation	-5658.39
Max deviation	165546.95
Min deviation	-661194.14
Standard deviation	48363.52

Table 14: Deviation Summary: LN-SPPSF Model + Geospatial

Metric	Value
Average sale price deviation	-10317.86
Max deviation	176927.00
Min deviation	-661706.88
Standard deviation	47482.40